THE ISLAMIA UNIVERSITY OF BAHAWALPUR

Department of Computer Science

FACIAL EMOTION DETECTION SYSTEM

Semester Project Report

Submitted in Partial Fulfillment of the Requirements of the Course

COMPUTER VISION

BS Computer Science – 6th Semester (3M)

Submitted To
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Submission Year: 2026

ABSTRACT

Facial expressions play an important role in expressing human emotions and are considered one of the most effective forms of non-verbal communication. The aim of this project is to develop a Facial Emotion Detection System capable of identifying emotions through facial expressions using both uploaded images and live webcam input.

The proposed system has been developed using Python and utilizes DeepFace and OpenCV libraries to analyze facial features and classify emotions. The application provides users with two input methods: image upload and webcam capture. The system detects faces, analyzes expressions, and predicts the dominant emotion along with its confidence score.

This project demonstrates the practical implementation of Computer Vision techniques and highlights the role of Artificial Intelligence in improving human-computer interaction.

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1. INTRODUCTION

Computer Vision is a branch of Artificial Intelligence that enables computers to interpret and understand visual information from images and videos. One of its emerging applications is facial emotion recognition, where machines identify human emotions by analyzing facial expressions.

Emotions such as happiness, sadness, anger, surprise, fear, disgust, and neutrality can provide valuable insights into human behavior. Facial Emotion Detection systems have applications in healthcare, education, security, customer service, and interactive systems.

This project focuses on designing a user-friendly application that detects emotions from uploaded images as well as real-time webcam input.

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2. PROBLEM STATEMENT

Humans can easily recognize emotions through facial expressions, but computers require intelligent algorithms to perform this task accurately.

The problem addressed in this project is:

How can a computer automatically detect and classify human emotions from facial images and webcam input using Computer Vision techniques?

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3. OBJECTIVES

- To develop a Facial Emotion Detection System using Python.
- To detect emotions from uploaded images.
- To perform real-time emotion detection using a webcam.
- To classify emotions accurately using DeepFace.
- To display the detected emotion along with confidence values.
- To provide a simple and user-friendly interface.

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4. SCOPE OF THE PROJECT

The scope of this project includes the development of a desktop-based application capable of recognizing emotions through facial expressions.

Included Features

- Emotion detection from uploaded images.
- Real-time webcam emotion recognition.
- Face detection and analysis.

• Display of confidence scores.
• Graphical user interface for user interaction.

Excluded Features
• Mobile application support.
• Voice emotion analysis.
• Cloud-based storage.
• Multi-camera integration.

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5. TOOLS AND TECHNOLOGIES USED

The following tools and technologies were used in the development of this project:

Technology	Purpose
Python	Programming Language
OpenCV	Face Detection and Image Processing
DeepFace	Emotion Analysis
TensorFlow	Deep Learning Backend
Tkinter	Graphical User Interface
Pillow	Image Handling
VS Code / Spyder	Development Environment

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6. METHODOLOGY

The system follows a sequence of steps to perform emotion detection.

Step 1: Input Acquisition
The user selects one of the following input methods: Upload an image from the computer, or Capture an image through the webcam.

Step 2: Face Detection
The system identifies the facial region within the input image using OpenCV.

Step 3: Emotion Analysis
DeepFace analyzes facial features and extracts emotional attributes.

Step 4: Emotion Classification
The dominant emotion is predicted from the following categories: Happy, Sad, Angry, Fear, Surprise, Disgust, Neutral.

Step 5: Result Display

The predicted emotion and confidence score are displayed to the user.

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7. SYSTEM WORKFLOW

The overall workflow of the proposed system is illustrated below:

User Input (Image Upload / Webcam)
↓
Face Detection (OpenCV)
↓
Emotion Analysis (DeepFace)
↓
Emotion Classification
↓
Result Display

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8. IMPLEMENTATION

The project was implemented using Python programming language. The application consists of the following modules:

Upload Module

This module allows users to browse and select an image from the local system for analysis.

Webcam Module

This module captures real-time images through the webcam and processes them for emotion recognition.

Emotion Detection Module

DeepFace is used to analyze facial expressions and identify the dominant emotion.

Output Module

The detected emotion and confidence percentage are displayed to the user.

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9. RESULTS AND DISCUSSION

The developed system successfully recognized emotions from both uploaded images and webcam captures. The application performed efficiently under normal lighting conditions and provided real-time results with acceptable accuracy.

The emotions recognized by the system include:

- Happy
- Neutral
- Sad
- Angry

- Fear
- Surprise
- Disgust

The system demonstrated that pre-trained deep learning models can effectively be integrated into desktop applications for practical Computer Vision tasks.

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10. ADVANTAGES

- Easy to use and understand.
- Supports both image upload and webcam input.
- Provides real-time emotion recognition.
- User-friendly graphical interface.
- Useful for educational purposes and research.
- Demonstrates practical implementation of Computer Vision concepts.

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11. LIMITATIONS

- Performance may decrease under poor lighting conditions.
- Accuracy depends on the quality of the input image.
- Extreme facial angles can affect detection accuracy.
- The system requires a clearly visible face.
- Results may vary for partially covered faces.

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12. FUTURE ENHANCEMENTS

- Age detection.
- Gender recognition.
- Attendance system integration.
- Emotion history database.
- Mobile application development.
- Cloud deployment.
- Multi-face tracking and analysis.

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13. CONCLUSION

The Facial Emotion Detection System was successfully developed using Python, OpenCV, and DeepFace. The system is capable of detecting and classifying emotions from uploaded images as well as live webcam input.

This project provided valuable practical experience in Computer Vision and Artificial Intelligence. It also demonstrated how deep learning techniques can be utilized to understand human emotions and improve human-computer interaction.

The successful completion of this project enhanced our knowledge of image processing, facial analysis, and real-world AI applications.

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14. REFERENCES

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2. OpenCV Team. OpenCV Documentation. Available at: https://docs.opencv.org/

3. TensorFlow Team. TensorFlow Documentation. Available at: https://www.tensorflow.org/

4. Sefik Ilkin Serengil. DeepFace Documentation. Available at: https://github.com/serengil/deepface

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15. GROUP CONTRIBUTION

Group Member	Contribution
Abdul Ghafar (Group Leader)	Project idea, system development, coding, integration, and testing
Muhammad Abdullah	Research work, report writing, and documentation
Rehan Ali Arman	User interface improvement, testing, and result analysis

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